

KERNEL MASTERS

Linux System Programming – Threads - Mini Project (Real Assignment)

WAP Producer thread and consumer thread synchronization using Pthreads and Signals? See the below Pseudo code.

Producer Thread Implementation:

```
void *producer(void *arg)
```

```
{
```

Step 1: Initialize producer signal handler.

Use signal (SIGUSR1,prod_sig) system call.

Step 2: Waiting for a signal from consumer thread inside infinite loop.

Use pause () system call inside infinite loop and wait for a signal from consumer thread.

```
while(1) {
```

```
    Pause();
```

```
}
```

```
}
```

Producer Signal Handler Implementation:

```
void pro_sig(int sig)
```

```
{
```

Step 1: Producer start produce the data until the buffer is FULL.

Use while () loop until Buffer is FULL.

Step 2: Whenever buffer is FULL raise a signal to consumer thread.

Use tkill () system call (or) pthread_kill library to raise a signal to consumer thread.

```
}
```

Consumer Thread Implementation:

```
void *consumer(void *arg)
```

```
{
```

Step 1: Initialize consumer signal handler.

Use signal (SIGUSR2,cons_sig) system call for signal handler initialization.

Step 2: Consumer raise a signal to producer thread.

Use tkill () system call (or) pthread_kill library to raise a signal to producer thread.

Step 3: Waiting for a signal from producer thread inside infinite loop.

Use pause () system call wait for a signal from producer thread.

```
while(1){
```

```
    pause();
```

```
}
```

```
}
```

Consumer Signal Handler Implementation:

```
void pro_sig(int sig)
```

```
{
```

Step 1: Whenever signal received from producer thread, start read the buffer until buffer is empty.

Use while () loop read Buffer until Buffer is EMPTY.

Step 2: Whenever the buffer is EMPTY, Consumer raise a signal to producer thread.

Use tkill () system call (or) pthread_kill library to raise a signal to producer thread.

```
}
```